

AWS EC2 + NGINX + GoDaddy Domain + SSL Configuration

2 Week – Day 6 Class

Topic

Hosting Website on AWS EC2 using NGINX with GoDaddy Domain and SSL Configuration

Architecture Flow

Browser



Domain Name (saikumar.online)



GoDaddy DNS



EC2 Public IP



Security Group



Ubuntu EC2



NGINX Server



index.html

Step 1 – Launch Ubuntu EC2 Server

Open: [AWS Console](#)

Go to:

- Services
 - EC2
 - Launch Instance
-

Configure EC2

Setting	Value
Name	Sai-Cloud-Nginx
AMI	Ubuntu Server 22.04
Instance Type	t2.micro
Key Pair	Create or Select Existing
Auto Assign Public IP	Enable

Step 2 – Configure Security Group

Important Security Group Rules

Port Purpose

22 SSH Login

80 HTTP Website Access

443 HTTPS Secure Website

Add Inbound Rules

Type	Port	Source
------	------	--------

SSH	22	Anywhere
-----	----	----------

HTTP	80	Anywhere
------	----	----------

HTTPS	443	Anywhere
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Step 3 – Connect to Ubuntu EC2

Copy:

- Public IP Address
-

SSH Command

`ssh -i mykey.pem ubuntu@PUBLIC-IP`

Example:

`ssh -i mykey.pem ubuntu@13.233.xx.xx`

Step 4 – Update Ubuntu Packages

Run:

`sudo apt update -y`

Purpose:

- Updates Ubuntu repositories.
-

Step 5 – Install NGINX

Run:

`sudo apt install nginx -y`

Purpose:

- Installs NGINX web server.
-

NGINX Installation Flow

Ubuntu Server



Install NGINX



Web Server Ready

Step 6 – Start NGINX Service

Run:

`sudo systemctl start nginx`

Step 7 – Enable NGINX Service

Run:

```
sudo systemctl enable nginx
```

Purpose:

- Automatically starts NGINX after reboot.

Step 8 – Verify NGINX Status

Run:

```
sudo systemctl status nginx
```

Expected Output:

active (running)

Step 9 – Open Website Directory

Run:

```
cd /var/www/html
```

Step 10 – Create Custom Web Page

Run:

```
sudo vi index.html
```

Press:

i

Paste Below HTML:

```
<html>
```

```
<head>
```

```
<title>Sai Cloud Services</title>
```

```
</head>
```

```
<body style="background-color:black; color:white; text-align:center; padding-top:200px;">
```

```
<h1>Welcome to Sai_Cloud_Services AWS Classes</h1>
```

```
<h2>NGINX Server Running Successfully</h2>
```

```
</body>
```

```
</html>
```

Save File:

ESC → :wq

Website Flow

Browser Request



NGINX Service



index.html



Website Displayed

Step 11 – Open Website in Browser

Copy:

- EC2 Public IP

Open Browser:

http://PUBLIC-IP

Example:

http://13.233.xx.xx

Expected Output

Welcome to Sai_Cloud_Services AWS Classes

Step 12 – Configure Domain in GoDaddy

Open: [GoDaddy](#)

Go to:

- My Products
 - DNS Management
-

Step 13 – Add DNS Records

Add Root Domain Record

Type Host Value

A @ EC2 Public IP

Add WWW Record

Type Host Value

A www EC2 Public IP

DNS Flow

Domain Name



GoDaddy DNS



EC2 Public IP



NGINX Website

Step 14 – Verify Domain

Wait:

- 5 to 15 Minutes

Open:

<http://saikumar.online>

and

<http://www.saikumar.online>

Step 15 – Install SSL Packages

Run:

```
sudo apt update
```

Install Certbot:

```
sudo apt install certbot python3-certbot-nginx -y
```

SSL Architecture

Browser



HTTPS Request



SSL Certificate



NGINX



Secure Website

Step 16 – Generate SSL Certificate

Run:

```
sudo certbot --nginx -d saikumar.online -d www.saikumar.online
```

During SSL Setup

Email Address

Enter your email.

Accept Terms

Type:

Y

Share Email

Type:

N

(Optional)

Redirect HTTP to HTTPS

Choose:

2

Purpose:

- Redirects HTTP traffic to HTTPS automatically.
-

Step 17 – Verify SSL

Open:

<https://saikumar.online>

Expected:

- Secure Lock Icon 
- HTTPS Enabled

- Secure NGINX Website

Step 18 – Verify Installed Certificates

Run:

```
sudo certbot certificates
```

Step 19 – Verify SSL Auto Renewal

Run:

```
sudo certbot renew --dry-run
```

Final Architecture

Browser



HTTPS Request



GoDaddy Domain



DNS Resolution



EC2 Public IP



Security Group



Ubuntu EC2



NGINX



index.html

Real-Time Interview Questions

Q1. What is EC2?

Answer:

- Amazon Web Services EC2 is a virtual server service in AWS cloud.
-

Q2. Why do we use Security Groups?

Answer:

- Security Groups control inbound and outbound traffic.
-

Q3. Why did you open Port 22?

Answer:

- Port 22 is used for SSH remote login.
-

Q4. Why did you open Port 80?

Answer:

- Port 80 allows HTTP website traffic.
-

Q5. Why did you open Port 443?

Answer:

- Port 443 allows HTTPS secure communication.
-

Q6. What is NGINX?

Answer:

- NGINX is a web server and reverse proxy server.
-

Q7. What is DNS?

Answer:

- DNS converts domain names into IP addresses.

Q8. Why do we add A Records in GoDaddy?

Answer:

- To map the domain with EC2 Public IP.
-

Q9. What is SSL?

Answer:

- SSL encrypts communication between browser and server.
-

Q10. Which tool is used for SSL generation?

Answer:

- Certbot Certbot.
-

Q11. How do you verify NGINX configuration?

`sudo nginx -t`

Q12. Website not opening after domain mapping. What will you check?

Answer:

1. Security Group
 2. DNS Records
 3. NGINX Status
 4. Public IP
 5. Firewall
 6. DNS Propagation
-

Q13. Explain the complete project flow.

Browser



GoDaddy Domain



DNS Resolution



AWS EC2 Public IP



Security Group



Ubuntu EC2



NGINX



Website Response

Sai Cloud Services